ECON2043 Experimental and Behavioural Economics
Seminar 1: Problem Set (Experiments)

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About the Seminar Series

Welcome to joining us!

Seminar series structure

- We have 6 seminars in total (w25, w27-w31).
- Primary focus: designing your experiment (group project).[1]
- Secondary focus: selected PS discussion.[2]

[1] Based on past experience, we will complete all experiments in w34, and use w35 for presentations. Further information about the group project: Moodle.

[2] Due to time constraints, we will not cover every PS question. Since solutions are available, repeating every solution step-by-step may have limited additional value.

Question 1: Paper Discussion

Read and discuss: [Fischbacher and Föllmi-Heusi, 2013]

- What is the research question?
- What is the experimental design?
- What do they find?

Then: think of one simple research question that can be tested with their setup.

Q1 Answer (1): Design and Main Findings

Research question

- How and why do people cheat when individual lying cannot be detected?

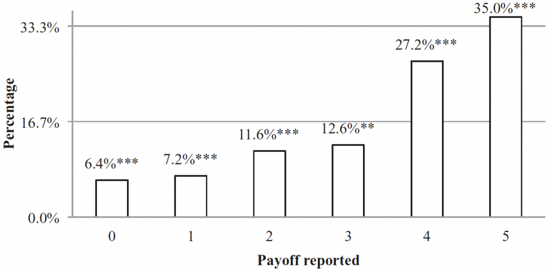
Design (simple)

- Private die roll; report payoff.
- If everyone is honest: each report should be about 1/6.

Payoff rule

Reported number	1	2	3	4	5	6
Payoff (CHF)	1	2	3	4	5	0

Q1 Answer (2): Results Snapshot



Types in data

- Fully honest: report real outcome.$(\hat{h}_{honest} \approx \frac{p_0}{1/6} = \frac{6.4\%}{1/6} \approx 38.4\% \approx 39\%)$
- Full lying: report maximum payoff. $(\hat{h}_{full\ lie} \approx \frac{p_5 - 1/6}{5/6} = \frac{35.0\% - 16.7\%}{5/6} \approx 22.0\%)$
- Partial lying: lie, but not to the maximum.$(\hat{h}_{partial} \approx p_4 - 1/6 = 27.2\% - 16.7\% = 10.5\%)$

Q1 Answer (3): Interpretation and Follow-up

Why partial lying? (possible reasons)

- Keep a positive self-image: “I am not fully dishonest.”
- Avoid appearing too greedy (report 4 instead of 5).
- Moral cost and money gain are balanced.

Follow-up question

- How can we apply this framework to a new research question?

Q1 Answer (4): What We Can Do Next

One simple example

- Research question: Does time pressure increase full lying?

Simple experiment

- Group A: no time limit before reporting.
- Group B: report within 5 seconds.
- Compare distributions of reported outcomes (especially 4 and 5).

Question 2

Think of examples from your life where you have run your own experiments and which:

- (a) were suitable experiments, or
- (b) violated the random assignment assumption.

Examples and Hints

Examples:

- The impact of weather on learning efficiency.
- The impact of listening to music on learning efficiency.

Have a look at last year's report.

Within-subject design vs. between-subject design

- Within-subject: observe each person under both treatment and control conditions.
- Between-subject (standard): one-shot design, each person is observed once in either treatment or control.

How to randomly assign treatment and control groups?

Example Random Assignment

You may use:

- Student ID[3]
- Birthdate

You cannot use:

- Gender
- Academic performance

Rule of thumb: use exogenous variables (not caused by ability/effort) for randomization, and avoid endogenous variables that are related to outcomes.

[3] Assign treatment if the last digit is odd, and control if it is even.

Question 3

In the lecture, we discussed an experiment that tests the effect of coffee on productivity. If you were in charge of running this experiment, how would you implement it specifically?

Outline two ways and discuss their advantages and disadvantages.

Hints

Lab experiment vs. field experiment

- What are the advantages and disadvantages of each?

An interesting thought experiment:

> *What would the ideal experiment look like, one with zero practical implementation constraints?*

Reference Answer: A Pure Lab Experiment

Treatment: a cup of coffee.
Control: a cup of caffeine-free coffee.

Choose outcome variables observable in lab that match the research question (productivity).

Advantage: full control over whether subjects drink coffee.
Disadvantage: lab is artificial; coffee consumption context differs from real life.

Think Further

Why do we let the control group have caffeine-free coffee, rather than another caffeine-free drink (e.g., soda)?

Hints

This design accounts for placebo effects:

- People may believe coffee increases productivity.
- People may believe no coffee decreases productivity.

These beliefs can directly affect effort and outcomes.

What can we do if we want to test placebo effects explicitly?

Reference Answer: A Real World Experiment (Field)

Treatment: drink coffee for the next 7 days.
Control: do not drink coffee for the next 7 days.

Ask participants to sign in at 12:00 and 18:00 to record coffee consumption, and rate productivity on a subjective scale.

Advantage: real-life behavior and often longer horizon; high practical relevance.
Disadvantage: limited control over actual behavior.

Question 4

Suppose you are interested in finding the causal effect of 0, 1, or 2 cups of black coffee on productivity.

Sketch treatment and control groups. How would you write the econometric model? How would you compute the ATE(s)?

Answer: Econometric Model

Three groups:

- 0 cups: pure control
- 1 cup: treatment 1
- 2 cups: treatment 2

Let $D_{1i} = 1$ if participant i is in the 1-cup group, and $D_{2i} = 1$ if in the 2-cup group. The baseline is 0-cup group ($D_{1i} = D_{2i} = 0$).

$$y_i = a + bD_{1i} + cD_{2i} + e_i$$

Answer: ATEs

Difference between one and two cups is the difference between c and b.

Three ATEs:

$$ATE_{1,0} = \mathbb{E}[y_i \mid D_{1i} = 1, D_{2i} = 0] - \mathbb{E}[y_i \mid D_{1i} = 0, D_{2i} = 0] = b$$

$$ATE_{2,0} = \mathbb{E}[y_i \mid D_{1i} = 0, D_{2i} = 1] - \mathbb{E}[y_i \mid D_{1i} = 0, D_{2i} = 0] = c$$

$$ATE_{2,1} = \mathbb{E}[y_i \mid D_{1i} = 0, D_{2i} = 1] - \mathbb{E}[y_i \mid D_{1i} = 1, D_{2i} = 0] = c - b$$

References

Fischbacher, U. and Föllmi-Heusi, F. (2013).
Lies in disguise–an experimental study on cheating.
Journal of the European Economic Association, 11(3):525–547.